

JERICHO B. BAAL

ELECTRONICS ENGINEER & TECHNICIAN

Embedded Software Engineer



Dasmariñas City, Cavite | jerichobaal1730@gmail.com | 0927 349 0769 | jerichobaal-portfolio.vercel.app

EDUCATION

Bachelor of Science in Electronics and Communications Engineering | 2015 - 2020

- Polytechnic University of the Philippines - Sta. Mesa, Manila
- President's List A.Y. 2019-2020
- Department of Science and Technology (DOST - SEI) - R.A. 10612 Scholar, 2017 - 2020

LICENCE AND CERTIFICATIONS

- Electronics Engineer 0078418 - Professional Regulatory Commission (2022)
- Electronics Technician 0025290 - Professional Regulatory Commission (2022)
- Data Science Essentials With Python - A Cisco Networking Academy program by DICT (2025)
- Full Stack Web Development Bootcamp - UDEMY , Dr. Angela Yu (2026)

PROFESSIONAL EXPERIENCE

Embedded Software Engineer - Thales Technologies Philippines Inc. | June 2026 - Present

- Developed Java card applets for GSM SIM microchips using C/C++, JAVA, VB.Net
- Evaluated new architecture and memory design for GSM SIM applets
- Wrote and reviewed the technical design specifications of secured software in C and Java
- Participated in integration testing of smart cards with mobile and servers

Special Science Teacher I - DepEd Bacoor City | October 2022 - May 2026

- Deliver robotics training covering Arduino/ESP32 C programming, sensor interfacing, GPIO control
- Engineered competitive robotics systems from concept to deployment for National Competitions
- Developed student IoT research projects and simulation research using modeling tools and LLMs
- Innovated classroom teaching for Earth Science, Biology, Chemistry, and Physics

Design Engineer - DENSO Philippines | January 2021 - September 2022

- Validated vehicular smart key system designs based on engineering drawing specifications
- Conducted laboratory tests on RF parameters (frequency, amplitude) using anechoic chamber
- Used Siemens NX for engineering drawing major or minor changes, assessing functional impact
- Performed benchmarking and monthly technical reporting for new RF electronic products

TECHNICAL SKILLS

Embedded Systems	Arduino, ESP32, GPIO, ADC, UART, I2C, IoT development, STM32
Programming	C, C++, Python, VB .NET, Java
Electronics	Schematic & PCB design (KiCad), circuit debugging and testing
Engineering Tools	Siemens NX, Git, VS Studio Code, LLMs, Jira, Confluence
Web Development	HTML, CSS, JavaScript, Node.js, Express.js, React.js, PostgreSQL

ACHIEVEMENTS & PROJECTS

- **1st Place, 1kg and 3kg Sumobot RC - National Science Quest 2025** Association of Science Educators in the Philippines, Teacher's Camp, Baguio City
- **2nd Place, Division Science and Technology 2025 Fair (Bacoor City)** - Mathematics and Computational Sciences - Comparative SUMO Simulation of Default and "Buhos" Traffic Control Schemes Across Varying Demand Levels at Selected Junctions and Intersections in Bacoor City, Cavite: Traffic Performance Analysis using Python"

Character reference is available upon request